

Elements of Statistics and Probability

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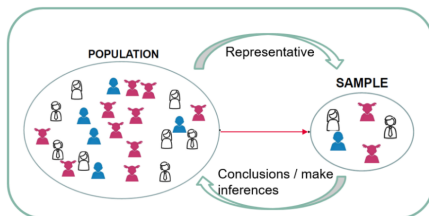
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Brief Review: Two Approaches in Statistics

- **Descriptive Statistics:** consists of methods for organising, displaying, and describing data by using tables, graphs, and summary measures
- **Inferential Statistics:** the procedure by which we reach a conclusion about a **population** on the basis of the information contained in a **sample** drawn from that population
 - **Population:** is the collection of all individuals or items under consideration in a statistical study
 - **Sample:** a representative part of the population from which information is collected
 - **Sampling:** the process of drawing a random sample
 - **Target population:** the population about which one wishes to make a conclusive statement (inference)

Population and Sample



- **Parameter of interest:** we may wish to know
 - the average final exam marks in STA201 for section 19
 - the average CGPA of EEE students in BRAC University
 - difference in the blood pressure between patient with glaucoma and healthy patient

Inferential Statistics

- **Interval estimation:** An interval estimate consists of two numerical values defining a range of values that, with a specified degree of confidence, most likely includes the parameter being estimated
- **Test of hypothesis:** the purpose of hypothesis testing is to aid the clinician, researcher, or administrator in reaching a conclusion concerning a population by examining a sample from that population
- An interval estimate can be used in hypothesis testing

Hypothesis: Definition & Concept

- **Hypothesis:** A hypothesis may be defined simply as a statement about one or more populations
- **Research hypothesis:** is the conjecture or supposition that motivates the research
 - It may be the result of years of observation on the part of the researcher
 - A public health nurse, for example, may have noted that certain clients responded more readily to a particular type of health education programme
- **Statistical hypothesis:** is a procedure to test whether a statement about a population (parameter of interest) is likely to be true or not (by chance) based on the sample data

Hypothesis: Definition & Concept

- A **test of hypotheses** or **significance test** is a procedure designed to assess the evidence provided by the data in favour of some statements, based on:
 - The null hypothesis, H_0 : statement of no effect, or no difference
 - The alternative hypothesis, H_a (or H_1): statement that there is a difference or an effect– what the experimenter would like to support

Test of Hypothesis: Examples

- Each of the following situations requires a significance test about a population mean. State the appropriate null hypothesis (H_0) and alternative hypothesis (H_a) in each case:

- (a) WHO concerned that the average weight of newborn in slum area is smaller than 2.5 Kg. Data on a random sample of 50 newborns from slum area is collected.

$$H_0 : \mu = 2.5$$

$$H_1 : \mu < 2.5$$

- (b) A study is conducted to test the hypothesis that people with glaucoma have higher-than-average blood pressure, where the average SBP for people of comparable age is 130 mm Hg.

$$H_0 : \mu = 130$$

$$H_1 : \mu > 130$$

Test of Hypothesis: Examples

- (c) If the machine is operating correctly, it produces bottles of mineral water with an average of 500 ml per bottle. The manager wants to investigate whether the machine is operating correctly before buying it and a random sample of 25 bottles were produced by the machine.

$$H_0 : \mu = 500$$

$$H_1 : \mu \neq 500$$

Basic Steps in Hypothesis Test

1. Determine null (H_0) and alternative (H_a) hypothesis
 - **Hypothesis** is a statement ABOUT a population parameter NOT ABOUT the sample statistic
2. Verify necessary data conditions (assumption), and if met, summarize the data into an appropriate **test statistics** (using appropriate data summary, or sample statistic)
3. Assuming the null hypothesis (H_0) is true, find either *rejection region (critical value)* or the *p-value* or *confidence interval*
4. Decide whether H_0 is rejected or not based on either *rejection region* or the *p-value* or *confidence interval*
5. Report the **conclusion** in the context of the problem (question of interest)

Test of Hypothesis: Some Definitions

- **Test Statistic:**

- is a random variable having a known probability distribution (under the null hypothesis, H_0)
- the test statistic serves as a decision maker, since the decision to reject or not to reject the null hypothesis depends on the magnitude of the test statistic
- An example of a test statistic:

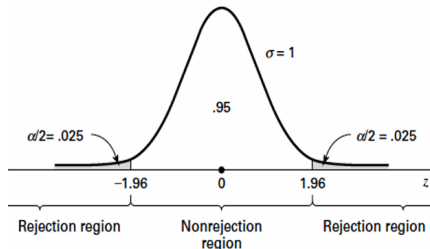
$$Z = \frac{\bar{X} - \mu_0}{SE(\bar{X})} = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$$

- General formula:

$$\text{Test statistic} = \frac{\text{Relevant statistic} - \text{Hypothesised parameter value under } H_0}{\text{SE of relevant statistic}}$$

Decision Rule

- All possible values that the test statistic can assume are points on the horizontal axis of the graph of the distribution of the test statistic and are divided into two groups; one group constitutes what is known as the **rejection region** or **critical region** and the other group makes up the **nonrejection region**

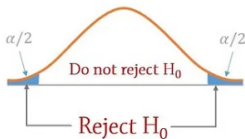


Decision Rule

Two-tailed

$$H_0: \mu = 23$$

$$H_1: \mu \neq 23$$

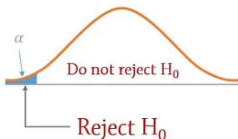


One-tailed

Left-tailed

$$H_0: \mu \geq 23$$

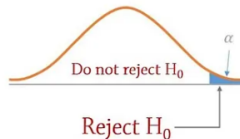
$$H_1: \mu < 23$$



Right-tailed

$$H_0: \mu \leq 23$$

$$H_1: \mu > 23$$



Decision Rule

- The *decision rule* tells us to reject the null hypothesis if the value of the test statistic that we compute from our sample is one of the values in the rejection region and to not reject the null hypothesis if the computed value of the test statistic is one of the values in the nonrejection region
- **The level of significance:** The level of significance, α , specifies the area under the curve of the distribution of the test statistic that is above the values on the horizontal axis constituting the rejection region
- The level of significance, α , is a probability and, in fact, is the probability of rejecting a true null hypothesis (rejecting the true null hypothesis is called “Type-I error”)

Test of Hypothesis: Decision and Error

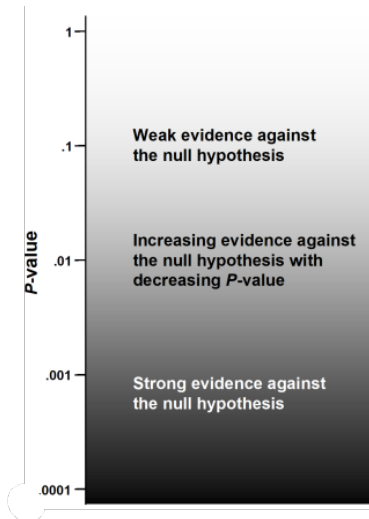
		Condition on null hypothesis	
		H_0 is true	H_0 is false
Decision/Action	Failed to reject H_0	correct decision	Type II error
	Reject H_0	Type I error	correct decision

- **Type I error:** occurs when the decision made is to 'reject' the true null hypothesis
- **Type II error:** occurs when the decision made is 'failed to reject' the false null hypothesis
- $\Pr(\text{Type I error}) = \alpha = \text{level of significance, usually we fixed it at 1\% or 5\%}$
- $\Pr(\text{Type II error}) = \beta$
- **Power** = $\Pr(\text{rejecting false } H_0) = 1 - \beta$

Decision based on p-value

- **p-value:** a p-value is the probability that the computed value of a test statistic is at least as extreme as a specified value of the test statistic if the null hypothesis is true
- A general rule worth remembering, then, is this: if $p\text{-value} \leq \alpha$, we reject the null hypothesis at $\alpha\%$ level of significance; if $p\text{-value} \geq \alpha$ we fail to reject the null hypothesis at $\alpha\%$ level of significance

Decision based on p-value



- The smaller the p-value, the lower the chance of getting a test statistic as extreme as the one observed if the null hypothesis is true
- Therefore the smaller the p-value, the stronger the evidence against the null hypothesis

Test for Single Population Mean: Known σ^2

- Let we want to test hypothesis

$$H_0 : \mu = \mu_0 \text{ against } H_a : \mu > \mu_0 \text{ (right-tailed test)}$$

- With the sample mean $\bar{x} = \sum x_i / n$ and assuming H_0 true, the test statistic

$$Z = \frac{\bar{x} - \mu_0}{SE(\bar{x})} = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} \sim N(0, 1)$$

- Critical value at α level of significance = $Z_{1-\alpha}$
 - Decision rule: if $Z_{cal} > Z_{1-\alpha}$ we reject H_0 (critical value approach)
 - Decision based on p-value: if $p\text{-value} = \Pr(Z \geq Z_{cal}) \leq \alpha$ we reject H_0
 - Decision based on CI: if $100(1 - \alpha)\%$ CI for $\mu \left(\bar{x} \pm Z_{1-\alpha} \frac{\sigma}{\sqrt{n}} \right)$ does not include the hypothesised value μ_0 we reject H_0

Test for Single Population Mean: Known σ^2

- For a **left-tailed test**

$$H_0 : \mu = \mu_0 \text{ against } H_a : \mu < \mu_0$$

- With the sample mean $\bar{x} = \sum x_i / n$ and assuming H_0 true, the test statistic

$$Z = \frac{\bar{x} - \mu_0}{SE(\bar{x})} = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} \sim N(0, 1)$$

- Critical value at α level of significance = Z_α
 - Decision rule: if $Z_{cal} < Z_\alpha$ we reject H_0 (**critical value approach**)
 - Decision based on p-value: if $p\text{-value} = \Pr(Z \leq Z_{cal}) < \alpha$ we reject H_0
 - Decision based on CI: if $100(1 - \alpha)\%$ CI for μ $\left(\bar{x} \pm Z_{1-\alpha} \frac{\sigma}{\sqrt{n}} \right)$ does not include the hypothesised value μ_0 we reject H_0

Test for Single Population Mean: Known σ^2

- For a **two-tailed test**

$$H_0 : \mu = \mu_0 \text{ against } H_a : \mu \neq \mu_0$$

- With the sample mean $\bar{x} = \sum x_i / n$ and assuming H_0 true, the test statistic

$$Z = \frac{\bar{x} - \mu_0}{SE(\bar{x})} = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} \sim N(0, 1)$$

- Critical value at α level of significance = $Z_{1-\frac{\alpha}{2}}$
 - Decision rule: if $Z_{cal} > Z_{1-\frac{\alpha}{2}}$ or $Z_{cal} < Z_{\frac{\alpha}{2}}$ we reject H_0 (**critical value approach**)
 - Decision based on p-value: if $p\text{-value} = 2 \Pr(Z \geq |Z_{cal}|) < \alpha$ we reject H_0
 - Decision based on CI: if $100(1 - \alpha)\%$ CI for $\mu \left(\bar{x} \pm Z_{1-\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \right)$ does not include the hypothesised value μ_0 we reject H_0

Seven Key Steps of Hypothesis Testing

- (1) Setting up the null and alternative hypothesis
- (2) Choosing the level of significance α
- (3) Choosing an appropriate test statistic
- (4) Setting up the decision rule
- (5) Calculating the test statistic chosen in step-3 from the observed sample
- (6) Making a decision based on Step-4 and Step-5
- (7) Interpreting the decision taken in step-6

Example 1

- An online medicine shop aims to enhance its delivery time. A sample of 49 orders was analyzed, revealing a mean delivery time of 100 minutes. Previous research indicates that the standard deviation of delivery time is 30 minutes. The medicine shop asserts that the mean delivery time for medicines is less than 120 minutes.
(a) Conduct a hypothesis test for the shop's claim at a significance level of 5%.

Example 1

- **Solution:**

- **Step-1:** Let μ be the mean delivery time for medicines. The the null and alternative hypotheses can be written as:

$$H_0 : \mu = 120$$

$$H_1 : \mu < 120$$

- **Step-2:** Consider the level of significance $\alpha = 0.05$
- **Step-3:** Since the population standard deviation (σ) is given, a Z test statistic would be appropriate. That is,

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$$

- **Step-4:** Decision rule: Reject H_0 if

$$Z_{cal} < Z_{\alpha} = Z_{0.05} = -1.64 \text{ (from the z table)}$$

Example 1

- **Step-5:** Given $\bar{x} = 100, \sigma = 30, n = 49$, the test statistic can be calculated as

$$Z_{cal} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{100 - 120}{30/\sqrt{49}} = -4.99$$

- **Step-6:** Since $Z_{cal} = -4.99 < Z_{\alpha} = -1.64$, the data provide enough evidence to reject H_0 at 5% significance level
- **Step-7:** Therefore, the data provide enough evidence to conclude that the mean delivery time for medicines is less than 120 minutes at 5% significance level

Exercise 1

- According to the American health association, a pregnant woman's average blood pressure is 120 mm Hg with standard deviation 4.5 mm Hg. To investigate the claim from the association, a research team has collected a random sample of 20 pregnant women and found the mean blood pressure of 125 mm Hg. Does the data provide any evidence of different mean blood pressure from accepted standard blood pressure. Use $\alpha = 0.01$.

Exercise 2

- The mean serum-creatinine level measured in 12 patients 24 hours after they received a newly proposed antibiotic was 1.2 mg/dL. If the mean and standard deviation of serum creatinine in the general population are 1.0 and 0.4 mg/dL, respectively, then, using a significance level of .05, test whether the mean serum-creatinine level in this group is different from that of the general population.